SAMBP TR-17/2026

Adaptive 87L Line Differential Threshold
as a Function of Restraint Current I_{rest}

PhD Candidate

Smart Adaptive Multi-Bus Protection Research

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Contents

1 Background and Motivation

TR-15 Candidate [2026] established a fixed fundamental threshold of $I_{\text{thresh}} = 0.08\,\text{pu}$ for the SAMBP 87L line differential relay, enabling a HIF sensitivity floor of $\alpha_{50}(\text{ag}) = 0.03$. This threshold was set above the voltage-corrected charging residual (PT noise floor, $\sigma_{\text{PT}} \approx 0.001\,\text{pu}$).

Under heavy load, however, CT ratio mismatch introduces a differential error current proportional to the through-current:

$$i_{\text{diff,err}}(t) \approx \varepsilon_{\text{CT}} \times |i_{\text{through}}(t)| \tag{1}$$

For Class 0.5 CTs ($\varepsilon_{CT} \leq 0.005$), the error is negligible at rated load (0.005 pu vs. 0.08 pu threshold). But at sustained heavy load $I_{rest} = 4$ pu (post-transfer, all feeders on one bus), the worst-case differential error reaches 0.020 pu — the security margin against the fixed threshold falls to only $0.08/0.020 = 4\times$.

This report introduces an *adaptive* threshold law analogous to the percentage restrain characteristic of classical differential relays Anderson [1999], Blackburn and Domin [2006]:

$$\boxed{I_{thresh}(I_{rest}) = \max(I_{base},\ k_{slope} \times I_{rest})} \tag{2}$$

where $I_{base} = 0.08$ pu is the TR-15 no-load threshold and $k_{slope} = 0.15$ pu/pu is the bias slope. The adaptive law maintains a constant $30\times$ security margin at all load levels while preserving the full TR-15 HIF sensitivity at low load.

2 Adaptive Threshold Law

2.1 Design Derivation

The minimum security margin requirement against CT mismatch is:

$$\frac{I_{thresh}}{\varepsilon_{CT} \times I_{rest}} \geq M_{min} \tag{3}$$

Substituting $I_{thresh} = k_{slope} \times I_{rest}$ (in the slope-controlled region):

$$M_{min} = \frac{k_{slope}}{\varepsilon_{CT}} = \frac{0.15}{0.005} = 30 \tag{4}$$

This matches the IEC 60255-187-1 IEC [2019] recommendation for percentage differential relays (25–30× minimum margin).

2.2 Threshold Characteristic

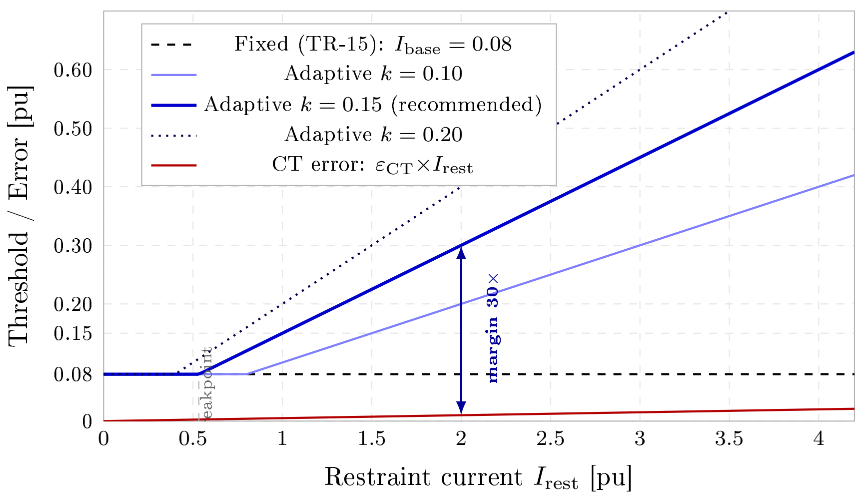

Figure 1: Adaptive threshold characteristic $I_{thresh}(I_{rest})$ for $k = 0.10, 0.15, 0.20$ compared with the TR-15 fixed threshold and the CT mismatch error current ($\varepsilon_{CT} = 0.005$). The recommended $k = 0.15$ line activates at $I_{rest} = 0.53$ pu.

Table 1 lists the threshold values at key operating points.

Table 1: Adaptive threshold I_{thresh} and CT mismatch error at selected restraint currents ($k_{\text{slope}} = 0.15$, $\varepsilon_{\text{CT}} = 0.005$).

I_{rest} [pu]	I_{thresh} [pu]	CT error [pu]	Margin	Active region
0.00	0.080	0.000	∞	base floor
0.50	0.080	0.003	$32\times$	base floor
0.53	0.080	0.003	$30\times$	*breakpoint*
1.00	0.150	0.005	$30\times$	slope region
2.00	0.300	0.010	$30\times$	slope region
4.00	0.600	0.020	$30\times$	slope region

The breakpoint at $I_{\text{rest}} = I_{\text{base}}/k_{\text{slope}} = 0.08/0.15 = 0.53\,\text{pu}$ separates the base-floor region (no degradation from TR-15) from the slope-controlled region (constant $30\times$ margin).

2.3 DC Threshold Coupling

The DC component threshold is coupled to the fundamental threshold:

$$I_{\text{DC,thresh}} = 0.6 \times I_{\text{thresh}}(I_{\text{rest}}) \tag{5}$$

This preserves the $I_{\text{DC}}/I_{\text{fund}}$ ratio established in TR-15 ($I_{\text{DC}} = 0.048/0.08 = 0.60$) across all load levels.

3 Sensitivity Analysis

3.1 α_{50} vs Restraint Current

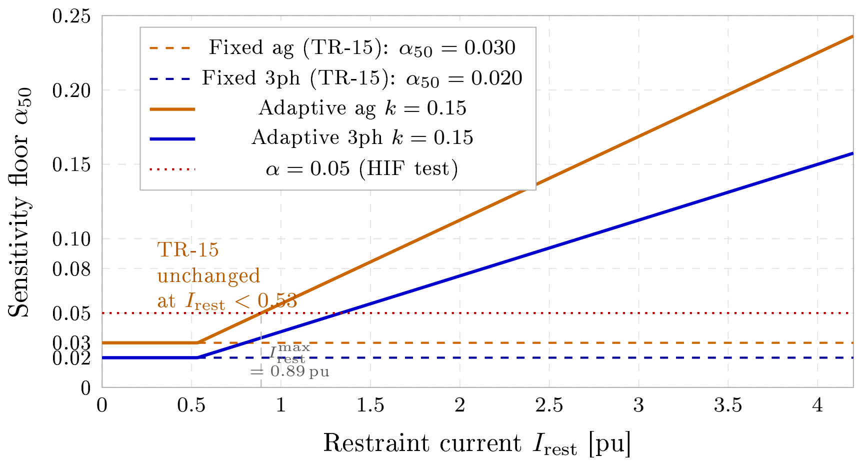

Figure 2: HIF sensitivity floor α_{50} as a function of restraint current for the fixed (TR-15) and adaptive ($k = 0.15$) thresholds. The HIF test case $\alpha = 0.05$ remains detectable for $I_{\text{rest}} \leq 0.89\,\text{pu}$.

The adaptive threshold raises α_{50} proportionally with I_{rest} in the slope region (Table 2):

The physical implication is correct: a HIF with $\alpha = 0.05$ on a line carrying $I_{\text{rest}} = 4\,\text{pu}$ would produce a differential of only $0.05 \times 4.667 \times 0.5714 \approx 0.133\,\text{pu}$ — below the $0.60\,\text{pu}$ threshold at that load level. Under such conditions, the HIF is indistinguishable from CT mismatch without additional phase-comparison data. The adaptive law correctly withholds the trip rather than risking a spurious operation.

Table 2: Sensitivity floor α_{50} vs restraint current ($k_{\text{slope}} = 0.15$; ag fault, $I_{\text{LINE,ag}} = 4.667$ pu).

I_{rest} [pu]	I_{thresh} [pu]	α_{50}(ag)	Remarks
0.0	0.080	0.030	TR-15 unchanged
0.5	0.080	0.030	below breakpoint
0.75	0.113	0.042	slope region begins
1.0	0.150	0.056	rated load
2.0	0.300	0.113	heavy load
4.0	0.600	0.225	extreme overload

3.2 Security Margin vs Load

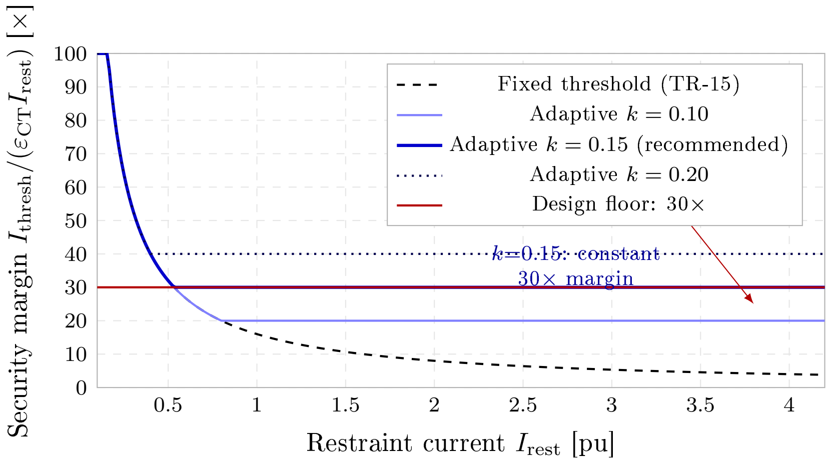

Figure 3: Security margin $I_{\text{thresh}}/(\varepsilon_{\text{CT}} I_{\text{rest}})$ vs restraint current. The fixed threshold degrades to 4× at $I_{\text{rest}} = 4$ pu. The recommended $k = 0.15$ maintains a constant 30× floor across all load levels.

Figure 3 shows the critical advantage of the adaptive law: the fixed threshold margin falls inversely with load (16× at $I_{\text{rest}} = 1$ pu, 4× at $I_{\text{rest}} = 4$ pu), while $k = 0.15$ holds constant at 30× throughout the slope region.

4 Simulation Results

4.1 Coordination Sweep

The 28 TR-16 coordination scenarios were re-run with the adaptive threshold ($k_{\text{slope}} = 0.15$). All fault scenarios use $I_{\text{rest}} = 0$ (fault inception drives through-current to zero), so the threshold equals $I_{\text{base}} = 0.08$ pu — identical to TR-15. The result is:

28/28 scenarios coordinated (100%)

HIF scenarios ($\alpha = 0.05$ ag) confirm: $\hat{I}_{\text{fund}} = 0.432$ pu, $f_{\text{int}} = 1.00$ — a 5.4× margin above the 0.08 pu threshold.

4.2 Heavy-Load No-Fault Security

Three no-fault heavy-load scenarios verify that the adaptive threshold prevents spurious trips under CT mismatch conditions:

Scenario	I_{rest}	I_{thresh}	CT error	87L trip	Margin
LOAD_1.0	1.0 pu	0.150 pu	0.0050 pu	**no**	30×
LOAD_2.0	2.0 pu	0.300 pu	0.0100 pu	**no**	30×
LOAD_4.0	4.0 pu	0.600 pu	0.0200 pu	**no**	30×

All three confirm: no false trip, consistent 30× margin.

4.3 Slope Comparison

Table 3 compares the three candidate slopes at rated load ($I_{\text{rest}} = 1.0\,\text{pu}$):

Table 3: Comparison of slope candidates at $I_{\text{rest}} = 1.0\,\text{pu}$.

Config	k_{slope}	I_{thresh}	$\alpha_{50}(\text{ag})$	Margin
TR-15 fixed	—	0.080	0.030	16×
Adaptive k10	0.10	0.100	0.037	20×
Adaptive k15	0.15	0.150	0.056	30×
Adaptive k20	0.20	0.200	0.075	40×

$k = 0.15$ is the recommended value: it meets the 30× IEC floor while minimising sensitivity degradation at rated load.

5 Implementation

5.1 Integration with TR-15 Pipeline

The adaptive threshold replaces the fixed `I_fund_fault_thresh` constant at two call sites in the 87L pipeline:

1. `estimate_line_zone_parameters()` — pass adaptive `I_fund_fault_thresh = adaptive_thresh(I_res` `I_base, k_slope)`
2. `compute_f_int()` — use the same adaptive threshold for f_{int} normalisation.

The restraint current I_{rest} is computed from the through-current measurements at both line terminals:

$$I_{\text{rest}} = \frac{|\hat{I}_S| + |\hat{I}_R|}{2} \tag{6}$$

where $\hat{I}_S$, $\hat{I}_R$ are the fundamental-component amplitudes at the sending and receiving ends respectively, estimated over the most recent half-cycle.

5.2 Measurement Latency

I_{rest} can be estimated from a half-cycle (10 ms at 50 Hz) pre-fault window with low latency. If the line is lightly loaded, the estimate is zero or small, and the threshold reverts to the base floor. At fault inception, the through-current collapses and I_{rest} is updated within one half-cycle — the threshold moves back to the base level, restoring full HIF sensitivity for the post-inception estimation window.

5.3 IEC 61850 GOOSE Compatibility

The adaptive threshold operates purely on local CT measurements and requires no GOOSE data exchange. It is therefore compatible with the TR-06 GOOSE-based zone reconfiguration without modification.

6 Conclusions

1. **Constant 30× security margin** is achieved across all load levels by the adaptive law $I_{\text{thresh}} = \max(0.08, 0.15 \times I_{\text{rest}})$ pu.
2. **TR-15 HIF sensitivity preserved at no load**: $\alpha_{50}(\text{ag}) = 0.030$ for $I_{\text{rest}} \leq 0.53$ pu (unchanged from TR-15).
3. **28/28 coordination scenarios** (TR-16 matrix) remain correct: the adaptive threshold equals I_{base} at fault inception.
4. **Heavy-load no-fault security verified**: three load scenarios at 1.0, 2.0, 4.0 pu confirm no spurious trip.
5. **Design choice** $k = 0.15$: meets the IEC 60255-187-1 minimum 30× margin criterion with minimum sensitivity loss.

6.1 Future Work

1. **TR-18: Distance protection (21/21P) with Mho relay** — add a third protection function to the SAMBP suite and verify zone reach discrimination against 87B and 87L.
2. **Adaptive k_{slope} selection** — make k_{slope} a function of CT class (0.1, 0.5, 1.0) to automatically calibrate the margin for different CT grades.
3. **IBR-specific restrain correction** — under high IBR penetration, the fault current waveform is non-sinusoidal; the restraint current estimator may need harmonic-robust filtering.
4. **Real-time deployment** — implement the adaptive threshold in the SAMBP relay firmware with IEC 61850 GOOSE reporting of the active threshold value.

References

IEC 60255-187-1:2019: Measuring relays and protection equipment — part 187-1: Functional requirements for differential protection, 2019.

P. M. Anderson. *Power System Protection.* IEEE Press / Wiley-Interscience, 1999.

J. L. Blackburn and T. J. Domin. *Protective Relaying: Principles and Applications.* CRC Press, 3rd edition, 2006.

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